

Newton's Divided Differences Interpolation Calculator

Numerical Methods PIT Project

Submitted by:

Kiert Jaderick L. Arquita – CpE2B

Submitted to:

Engr. Jodie Rey Fernandez

1. Project Architecture Overview

The Newton's Divided Differences Interpolation Calculator is a web-based educational application developed using Flask and Python. The system allows users to input interpolation data points, compute the divided difference table, generate the Newton interpolation polynomial, and visualize the interpolation curve.

System Components

Component	Description
Flask Backend	Handles routing, validation, interpolation computation, graph generation, and template rendering.
HTML Templates	Displays the calculator page, examples page, theory page, and result page.
CSS Styling	Provides the responsive user interface and visual layout.
Interpolation Module	Contains the Newton Divided Differences algorithms and polynomial generation logic.
Matplotlib	Generates interpolation graphs dynamically.
Vercel Deployment	Hosts the Flask web application online.

Application Flow

1. User inputs x-values, y-values, and interpolation target.
2. Flask validates and parses the inputs.
3. Newton Divided Difference calculations are performed.
4. Polynomial equation is generated.
5. Graph visualization is rendered.
6. Results are displayed on the result page.

2. Application Screenshots

Discussion / Theory Page

The screenshot shows a dark-themed web page titled "Mathematical Discussion" under the heading "NEWTON'S DIVIDED DIFFERENCES". It features a section titled "Newton vs. Lagrange Interpolation" with a sub-section "Lagrange Form" containing the formula
$$P_{n-1}(x) = \sum_{i=0}^{n-1} y_i \prod_{j=0, j \neq i}^{n-1} \frac{x - x_j}{x_i - x_j}$$
 and a note about its recomputation. Below is the "Newton's Divided Difference Form" with the formula
$$P_{n-1}(x) = \sum_{k=0}^{n-1} f[x_0, \dots, x_k] \omega_k(x), \quad \omega_k(x) = \prod_{j=0}^{k-1} (x - x_j)$$
 and a note about adding new points. A "Computational Complexity Comparison" table follows:

OPERATION	LAGRANGE	NEWTON (DD)
Initial setup	None	$O(n^2)$
Evaluate at one point	$O(n^2)$	$O(n)$
Add a new data point	$O(n^2)$	$O(n)$
Modify existing data	$O(n)$	$O(n^2)$

At the bottom, there are three buttons: "Same unique polynomial", "Different algebraic form", and "Newton favored for streaming data".

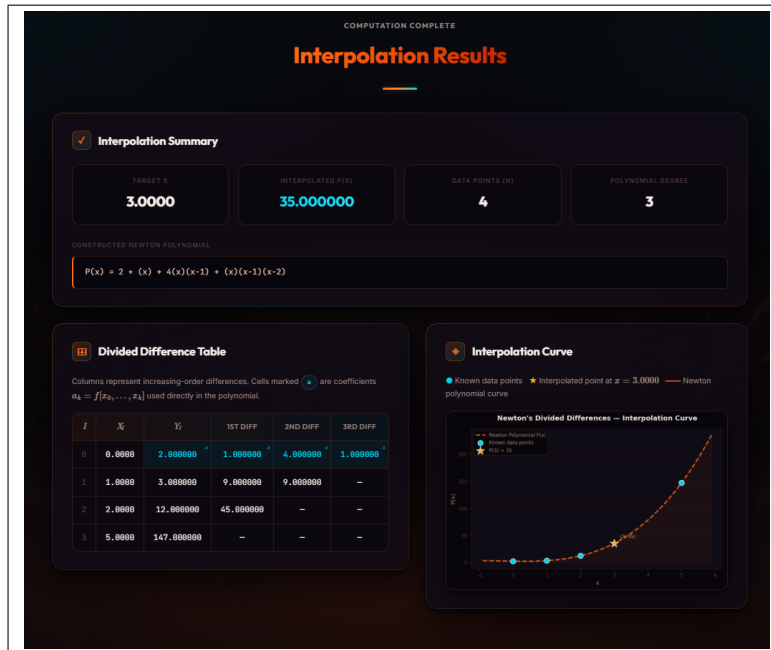
The theory page explains Newton's Divided Differences formulas and interpolation concepts.

Calculator Page

The screenshot shows a dark-themed web page titled "Interactive Calculator". It has three input fields: "X VALUES — COMMA SEPARATED" with a placeholder "e.g. 1, 2, 4", "Y VALUES — COMMA SEPARATED" with a placeholder "e.g. 3, 11, 47", and "INTERPOLATION TARGET (X)" with a placeholder "e.g. 3". Below the Y values field is a note: "Corresponding f(x) values — must match length of X." At the bottom, there are four buttons for "QUICK-LOAD DEMO DATASET": "Linear (n=2)", "Quadratic (n=3)", "Cubic (n=4)", and "Degree-4 (n=5)". A large orange button at the very bottom says "Calculate & Plot".

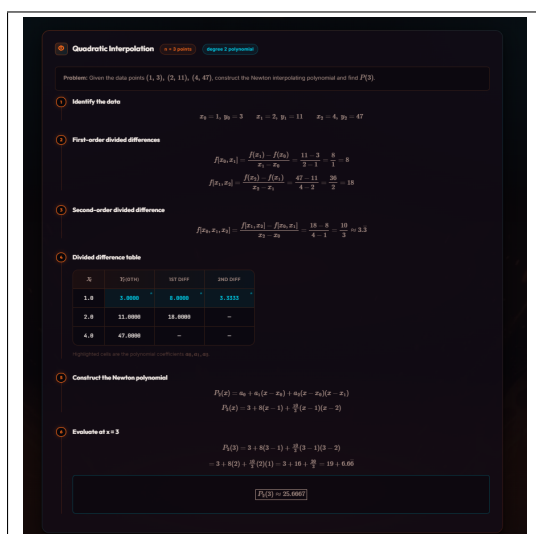
The calculator page accepts user interpolation inputs and computes the Newton polynomial.

Result Page



The result page displays the divided difference table, interpolation polynomial, and generated graph.

Worked Examples Page



Example problem demonstrating Quadratic Interpolation.



Example problem demonstrating Cubic Interpolation.

3. Explanation of Safe User Input Parsing

The application validates and safely parses all user inputs before calculations are performed.

Validation Steps

- Detect empty input fields
- Validate numeric conversion
- Ensure equal number of x-values and y-values
- Require at least 2 data points
- Reject duplicate x-values

Importance of Validation

Input validation helps prevent:

- Invalid calculations
- Application crashes
- Division-by-zero errors
- Incorrect interpolation results

Example Validation Logic

```
if len(x_values) != len(y_values):
    raise ValueError(
        "The number of X values must match Y values."
    )

if len(x_values) != len(set(x_values)):
    raise ValueError(
        "Duplicate X values are not allowed."
    )
```

4. Errors and Challenges Encountered

Problem / Challenge	Solution
Invalid numeric input	Implemented validation and try-except blocks for safer parsing.
Incorrect number of values	Added checks to ensure x-values and y-values have equal lengths.
Duplicate x-values	Added validation to avoid division-by-zero errors during interpolation.
Graph not displaying properly	Adjusted Matplotlib settings and image rendering logic.
Terminal deployment errors	Fixed missing dependencies and corrected project file structure.
CSS layout issues	Modified spacing and responsive layout styles in the stylesheet.
Broken page routes	Added Flask route handling and fallback redirects.
Missing image files	Verified correct image filenames and directory locations.
Incorrect Flask port configuration	Updated the application configuration for successful deployment.
Typing mistakes in code	Reviewed and corrected syntax and spelling errors in Python files.

5. Project Deployment Link

Vercel Deployment:

<https://nm-pit-7kvj.vercel.app>